

Sales Data Analysis Project

1. Project Overview

This project focuses on analyzing a retail sales dataset using Excel, SQL, and Power BI to generate meaningful business insights. The objective was to clean raw data, perform analysis, and build an interactive dashboard for better decision-making.

2. Tools & Technologies Used

- Microsoft Excel (Data Cleaning & Preparation)
 - SQL (Data Analysis)
 - Power BI (Data Visualization & Dashboarding)
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3. Dataset Description

The dataset contains over 5,000+ sales records with the following fields:

- Order ID
- Order Date
- Ship date
- Ship mode
- Segment
- State
- Country
- Market
- Region
- Product id
- Category
- Sub category
- Sales
- Quantity
- Discount
- Profit
- Shipping cost
- Order priority

4. Data Cleaning (Excel)

- Removed duplicates and handled missing values
- Standardized column formats (dates, text, numeric values)
- Ensured consistency in category and region naming

5. Data Analysis (SQL)

Performed various SQL queries to extract insights:

- Monthly sales trend analysis
- Region-wise sales and profit performance
- Category-wise revenue and profitability
- Customer purchase behavior

Sample SQL Queries Used:

1. Region-wise Sales and Profit

```
SELECT region,  
SUM(sales) AS total_sales,  
SUM(profit) AS total_profit  
FROM superstore GROUP BY region  
ORDER BY total_sales DESC;
```

2. Category-wise Performance

```
SELECT category,  
SUM(sales) AS total_sales,  
SUM(profit) AS total_profit  
FROM superstore  
GROUP BY category;
```

3. Profit Margin by Category

```
SELECT category,  
SUM(profit)/SUM(sales) * 100 AS profit_margin  
FROM superstore  
GROUP BY category;
```

SQL Concepts Used:

- SELECT, WHERE
- GROUP BY, ORDER BY
- Aggregate functions (SUM, COUNT)

6. Data Visualization (Power BI)

Developed an interactive dashboard including:

Key KPIs:

- Total Sales
- Total Profit
- Total Orders

Visualizations:

- Sales by Category (Bar Chart)
- Region-wise Performance (Map/Bar Chart)
- Profit Analysis Dashboard

Features:

- Dynamic filters (Region, Category)
 - Interactive slicers
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7. Key Insights

- Certain regions contributed significantly higher sales compared to others
 - Some product categories showed high sales but low profitability
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8. Business Impact

- Automated reporting through Power BI dashboard
 - Reduced manual analysis effort
 - Enabled faster and data-driven decision-making
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9. Conclusion

This project demonstrates the ability to work with end-to-end data analysis, from cleaning and processing to visualization and insight generation using industry-relevant tools.

10. Skills Demonstrated

- Data Cleaning
 - Data Analysis
 - SQL Querying
 - Dashboard Development
 - Business Insight Generation
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11. Recommendations

Based on the analysis, the following recommendations can help improve business performance:

- **Focus on high-performing regions:** Allocate more marketing budget and inventory to regions contributing the highest revenue.
 - **Improve low-profit categories:** Review pricing, discounts, and supplier costs for categories with low profit margins.
 - **Enhance customer targeting:** Create loyalty programs for repeat customers and personalized offers for high-value segments.
 - **Optimize reporting process:** Continue using automated dashboards to reduce manual effort and enable faster decision-making.
-